Multinomial Logistic Regression
SoftMax Regression.

Let's take placement dataset.

1 → Yes
0 → NO
2 → Opt out [didn't stay in the interview, coz of startup, or business or other reason].

- Logistic Regression is specialization of the softmax regression for binary output or two output [either 0 or 1]
  [class = 2]

Softmax Function.

$$\sigma(z)_i = \frac{e^{z_i}}{\sum_{j=1}^{k} e^{z_j}}$$

k ⇒ no. of classes
here 3 [Yes, NO, optout]

For Yes: i = 1

$$\sigma(z)_1 = \frac{e^{z_1}}{\sum_{j=1}^{3} e^{z_j}} = \frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

if then

~~No:~~

$$\cancel{\sigma(z)_0 = \frac{e^{z_0}}{\sum_{j=0}^{3} e^{z_j}} = \frac{e^{z_0}}{e^{z_0}+e^{z_2}+e^{z_3}}}$$

- It provides probability of each class.
  $0 < \sigma < 1$.

- $\sigma(z)_0 + \sigma(z)_1 + \sigma(z)_2 = 1$

- For class = 2 softmax $\rightarrow$ sigmoid f^n

Training Intuition

cgpa	iq	placement
		0
		1/2

Step 1: Do One Hot Encoding in output [placement].

cga	iq	0	1	2
		1	0	0
		0	1	0
		0	0	1

Page No.
Date: / /

above dataset will be divided in

Dataset

Dataset 0

cgpa	iq	0

Dataset 1

cgpa	iq	1

Dataset 2

cgpa	iq	2

model 0 model 1 model 2

the eqn: $w_0 + w_1 x_1 + w_2 x_2 = 0$.

After training there: we get w_0, w_1, w_2 of all the sets.

$w_0^{(0)}, w_1^{(0)}, w_2^{(0)}$ $w_0^{(1)}, w_1^{(1)}, w_2^{(1)}$ $w_0^{(2)}, w_1^{(2)}, w_2^{(2)}$

Find $z = w_0 + w_1 x_1 + w_2 x_2$ for each

For eg: model 0 : $z = w_0^{(0)} + 7 w_1^{(0)} + 70 w_2^{(0)}$
$\downarrow$ $\{7, 70\}$

Here we get

0 yes	0_x
1	0

Here

1_{yes}	1_x
1	0

Here

2 yes	2 x

$0_x \rightarrow$ won't be 0

$1_x \rightarrow$ won't be 1.

$2_x \rightarrow$ won't be 2

Page No.
Date: / /

$\sigma(No) = \frac{e^{z_1}}{e^{z_1}+e^{z_2}+e^{z_3}}$ $\sigma(Yes) = \frac{e^{z_2}}{e^{z_1}+e^{z_2}+e^{z_3}}$ $\sigma(\text{Opt Out}) = \frac{e^{z_3}}{e^{z_1}+e^{z_2}+e^{z_3}}$

for now: $= 0.4$ $= 0.35$ $= 0.25$.

Here His placement's won't be happen
coz. $0.4 > 0.35 > 0.25$.
$\sigma(No) > \sigma(Yes) > \sigma(\text{Opt Out})$

- For bigger dataset of higher classes, It will be slower.

Soft Max Loss Function

$$L = \frac{-1}{m} \sum Y_i \log(\hat{Y}_i) + (1-Y_i)\log(1-\hat{Y}_i)$$

For softmax: $L = \frac{-1}{m} \sum_{i=1}^{m} \sum_{k=1}^{k} \cancel{-y_k \log(\hat{y}_k)}\ y_k^{(i)} \cdot \log \hat{y}_k^{(i)}$